

■ Brain Tumor Classification

MobileNetV2 · Transfer Learning · Streamlit Deployment

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1. Project Overview

This project implements a deep learning pipeline to classify brain MRI scans into four diagnostic categories using **MobileNetV2** with transfer learning. The trained model is hosted on HuggingFace and served through an interactive Streamlit web application, making it accessible to researchers and educators without any local setup.

2. Classification Classes

Class	Description	Characteristics
■ Glioma	Tumor in glial cells	Most common malignant brain tumor
■ Meningioma	Tumor in meninges	Usually benign; slow-growing
■ No Tumor	No tumor detected	Healthy brain MRI scan
■ Pituitary	Tumor in pituitary gland	Affects hormonal regulation

3. Model Architecture

MobileNetV2 was chosen as the backbone due to its efficiency and strong performance on image classification tasks with limited compute. The architecture uses depthwise separable convolutions, significantly reducing parameters while maintaining accuracy.

Property	Value
Architecture	MobileNetV2 (Transfer Learning)
Pre-trained On	ImageNet
Input Shape	224 x 224 x 3
Output Classes	4
Framework	TensorFlow / Keras
Optimizer	Adam

Loss Function	Categorical Cross-Entropy
Model File	mobile_model.h5

4. Dataset

The model was trained on the **Brain Tumor MRI Dataset** from Kaggle, curated by Masoud Nickparvar. The dataset contains thousands of MRI images across four balanced classes, split into training and testing sets.

Source	Kaggle — Brain Tumor MRI Dataset
Classes	Glioma, Meningioma, No Tumor, Pituitary
Format	JPEG images
Split	Train / Test

5. Deployment

The application is deployed using **Streamlit Community Cloud**. The model is loaded on-demand from HuggingFace Hub, keeping the repository lightweight with no large binary files committed to Git.

Component	Technology
Web Framework	Streamlit
Model Hosting	HuggingFace Hub (inosukeo1/vg16_tumor)
Hosting Platform	Streamlit Community Cloud
Model Download	huggingface_hub.hf_hub_download()
Image Processing	Pillow (PIL)
Inference	TensorFlow / Keras

6. Run Locally

Clone the repository

```
git clone https://github.com/AmanShingane/brain-tumor.git
cd brain-tumor
```

Install dependencies

```
pip install -r requirements.txt
```

Launch the app

```
streamlit run app.py
```

The app opens at **http://localhost:8501**. The model is downloaded automatically from HuggingFace on the first run and cached locally.

7. Repository Structure

File / Folder	Purpose
<code>app.py</code>	Streamlit web application — main entry point
<code>brain_tumor.ipynb</code>	Model training, evaluation and visualisation notebook
<code>requirements.txt</code>	Python package dependencies
<code>README.md</code>	Project documentation

8. Disclaimer

■ ■ For Educational & Research Use Only

This tool is intended for educational and research purposes only. It should **not** be used as a substitute for professional medical diagnosis. Always consult a qualified medical professional for any health concerns.